BAY AREA LIGHT SOURCES JOINT USERS' MEETING

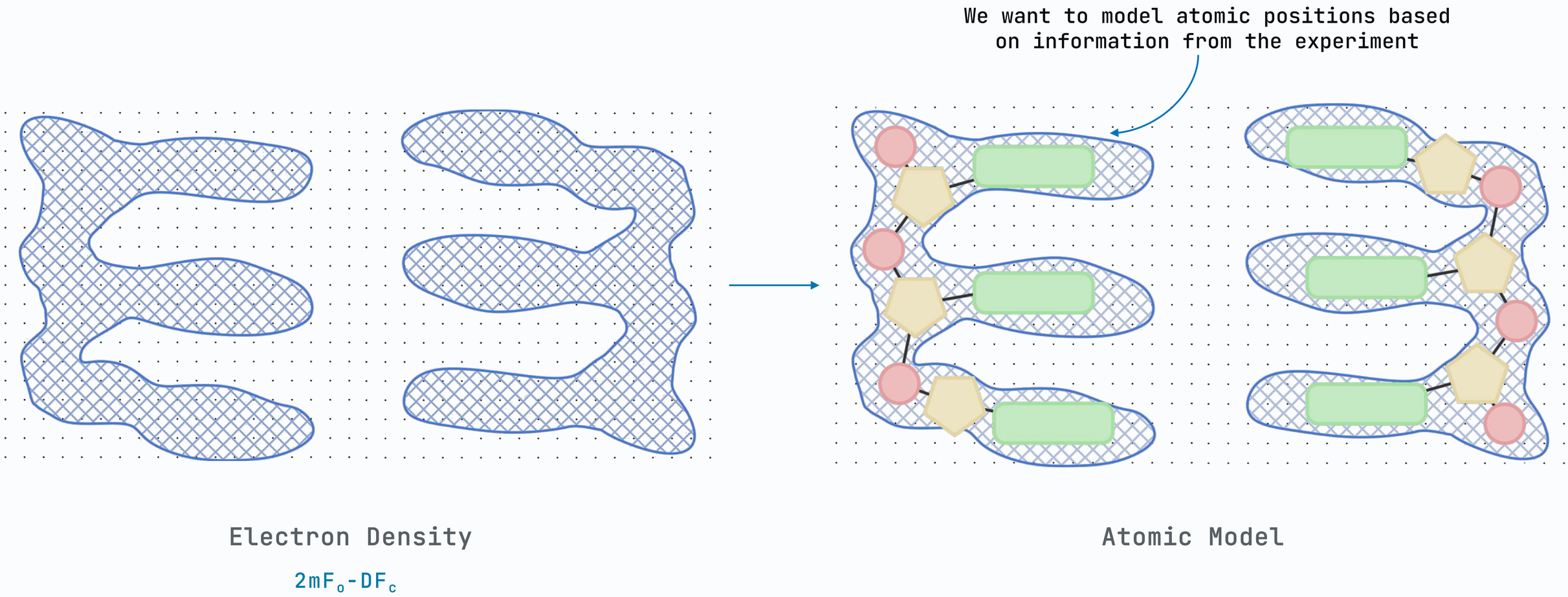


Automated nucleic acid model building using deep learning

Jordan Dialpuri, PhD - MRC Laboratory of Molecular Biology

25th September 2026

What is model building?



Schematic of automated model building.

What software packages help us do this?

Automated Model Building

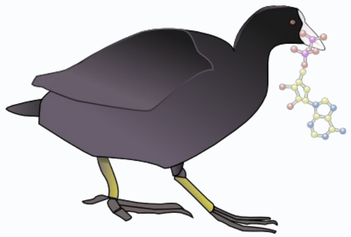


ModelCraft





Interactive Model Building



Coot



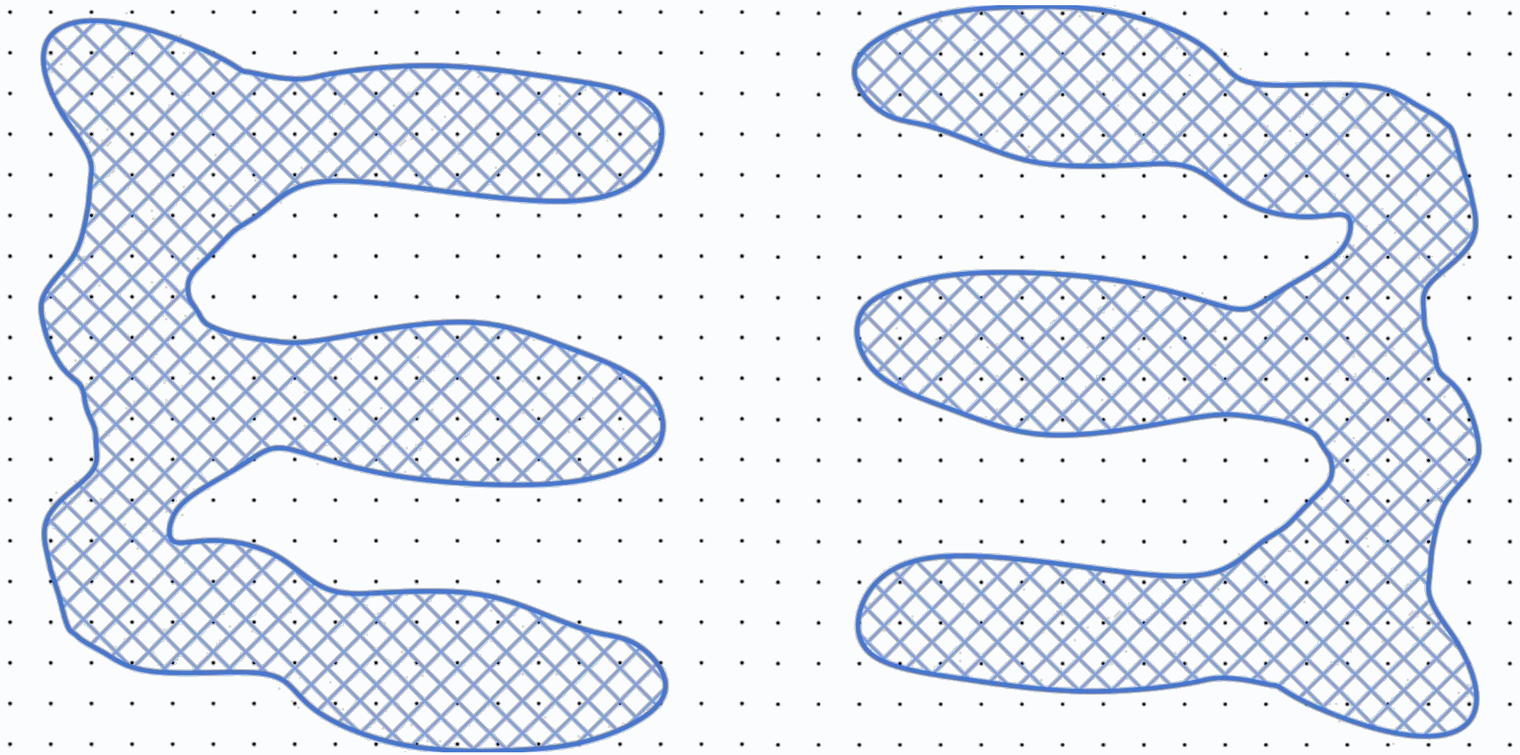
ChimeraX

It is often recommended to attempt to use automated model building software, first

And finish models interactively, because automated model building software packages aren’t perfect

What is the core idea behind automated model building programs?

Is there nucleic acid? If so, where?



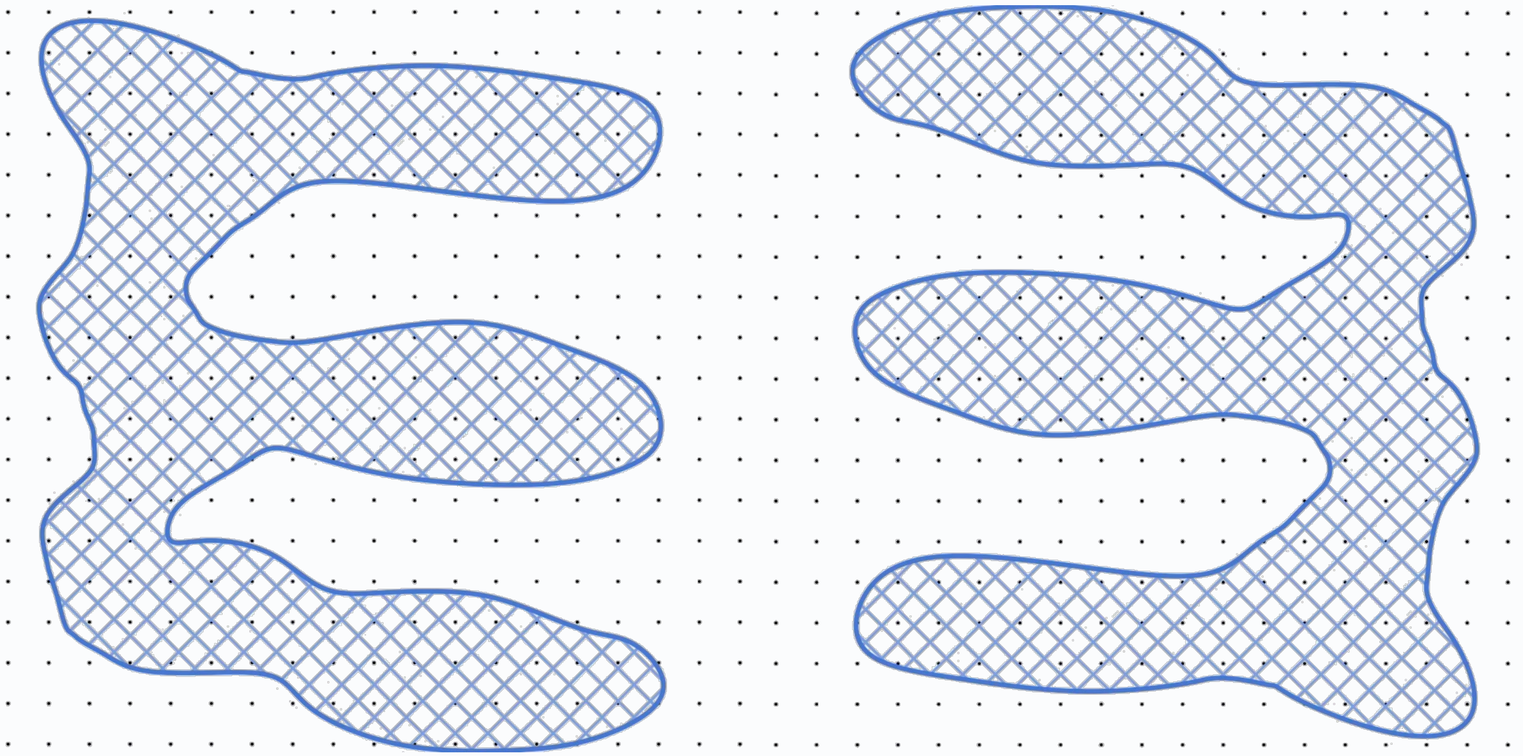
Electron Density

$2mF_o\text{-}DF_c$

What is the core idea behind automated model building programs?

Is there nucleic acid? If so, where?

Fingerprint identification



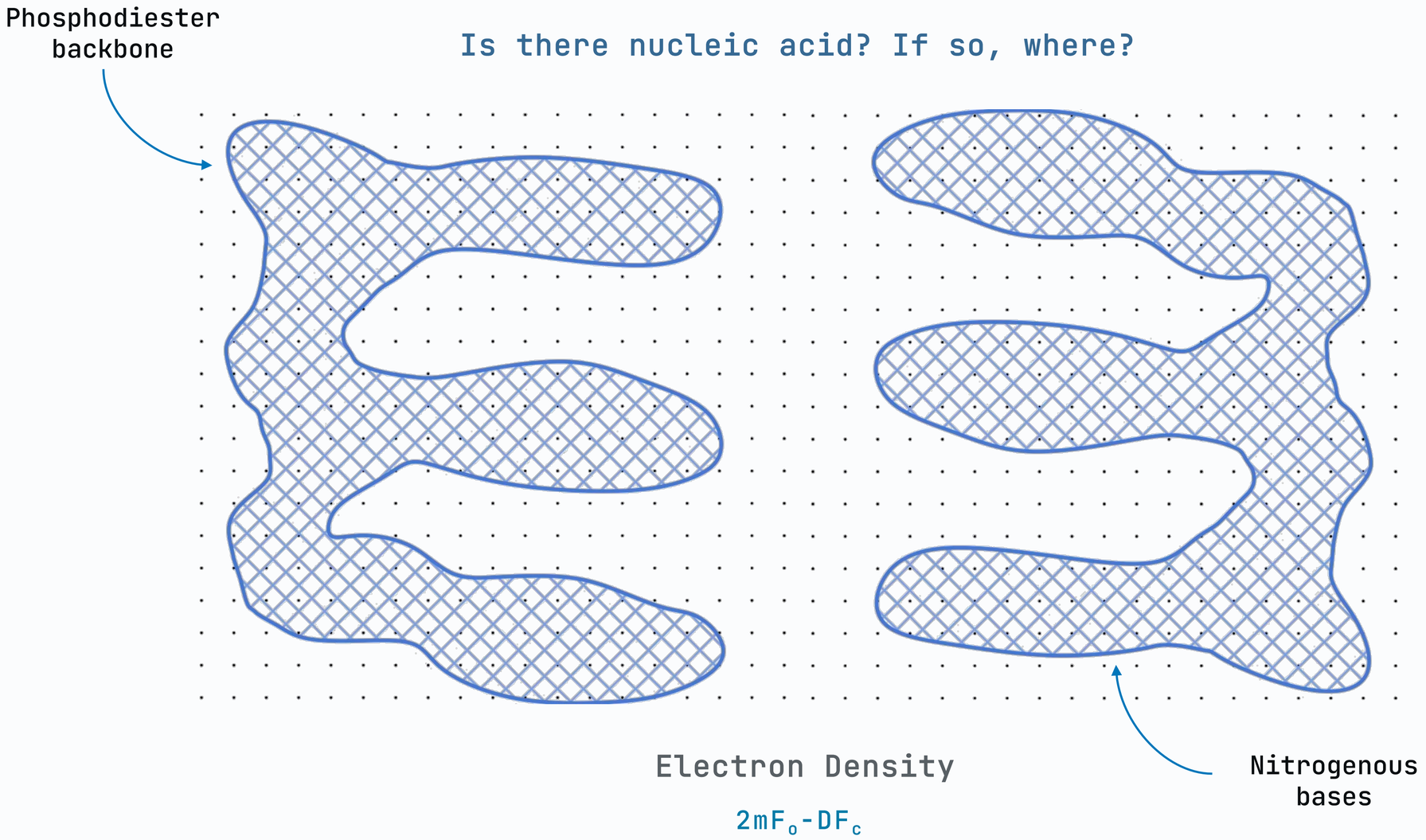
Free atom placement

Template fitting

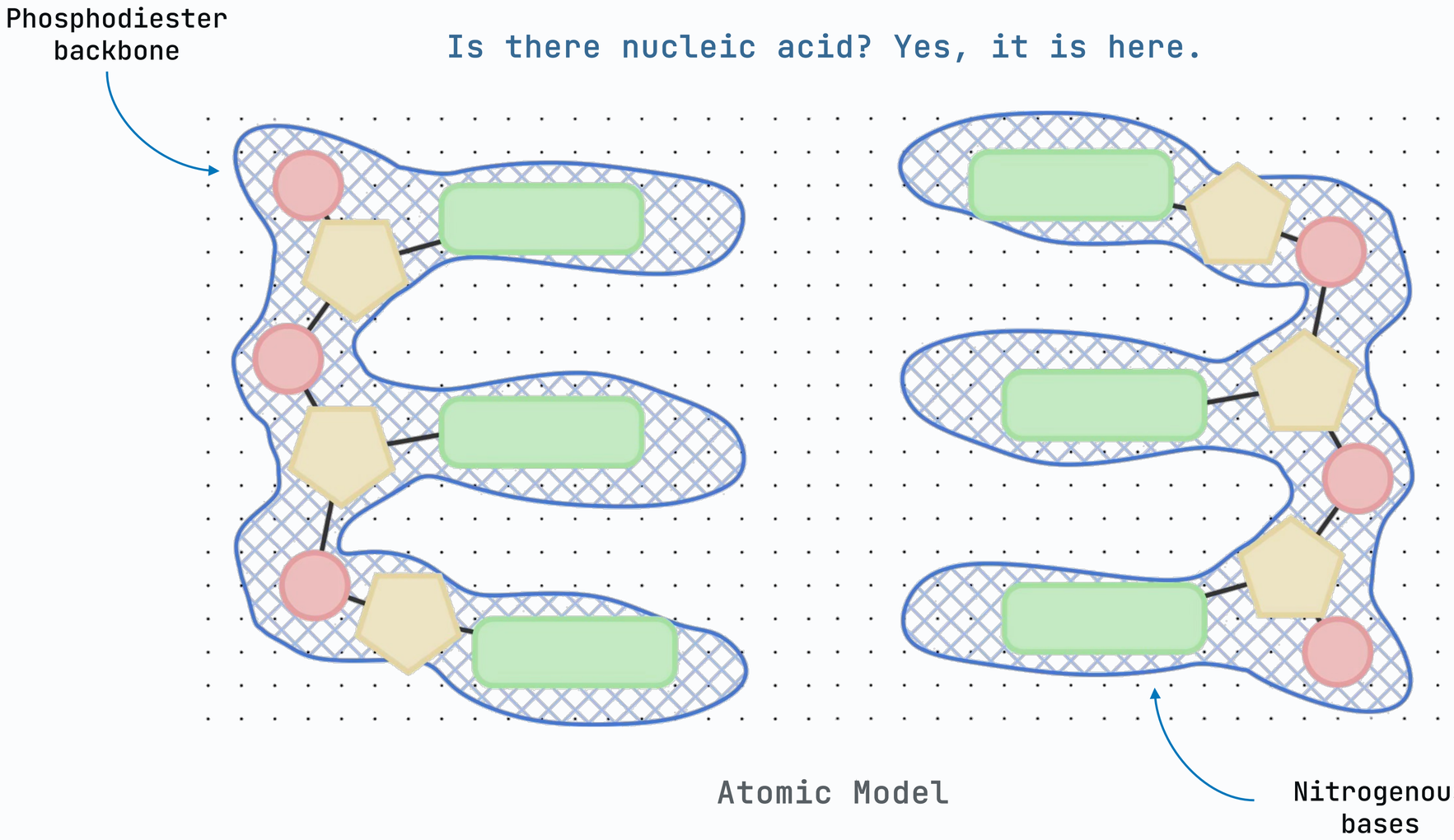
Electron Density

$2mF_o-DF_c$

What is the core idea behind automated model building programs?



Once software methods locate features, they can build a model





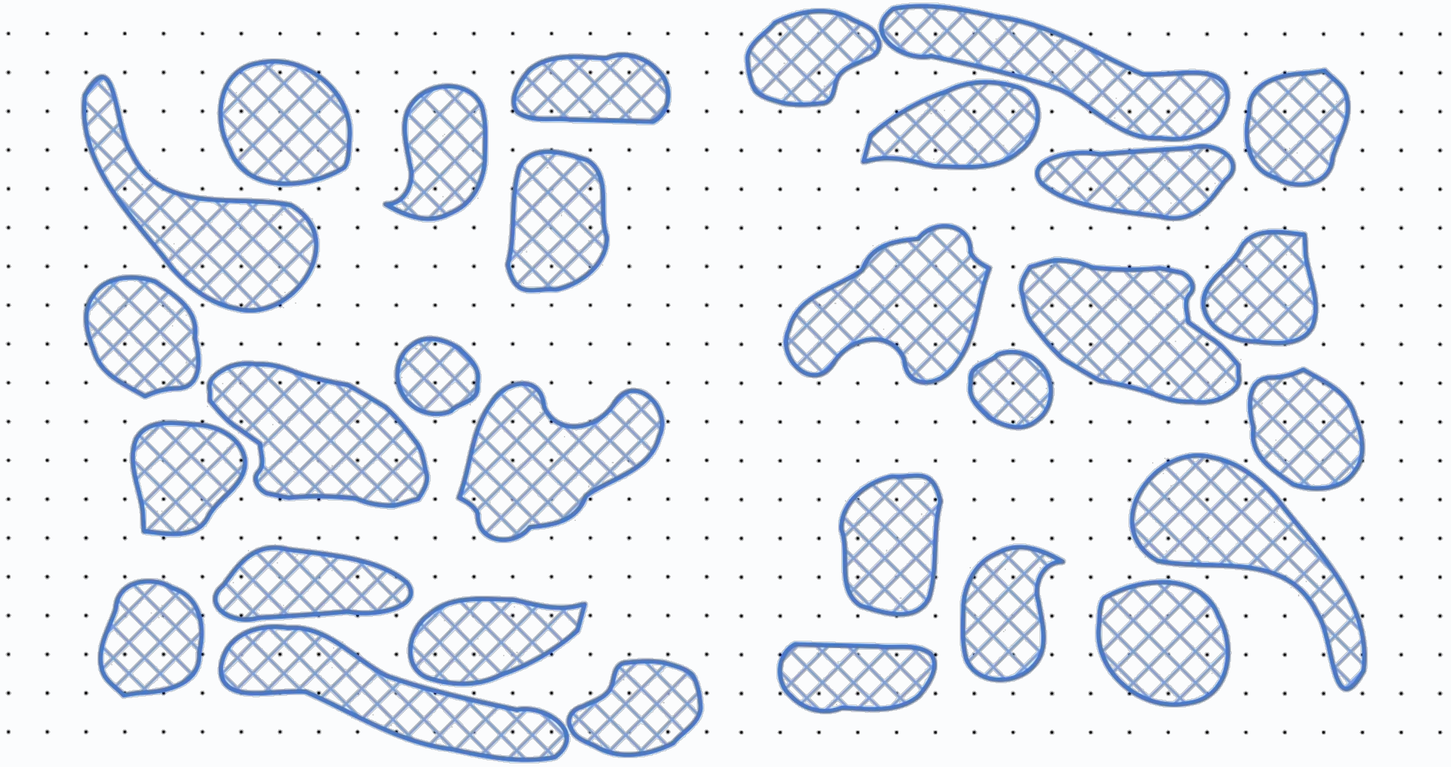




Do these methods work well in more difficult cases?

Phosphodiester backbone ?

Is there nucleic acid? If so, where?



Electron Density

$2mF_o$-DF_c

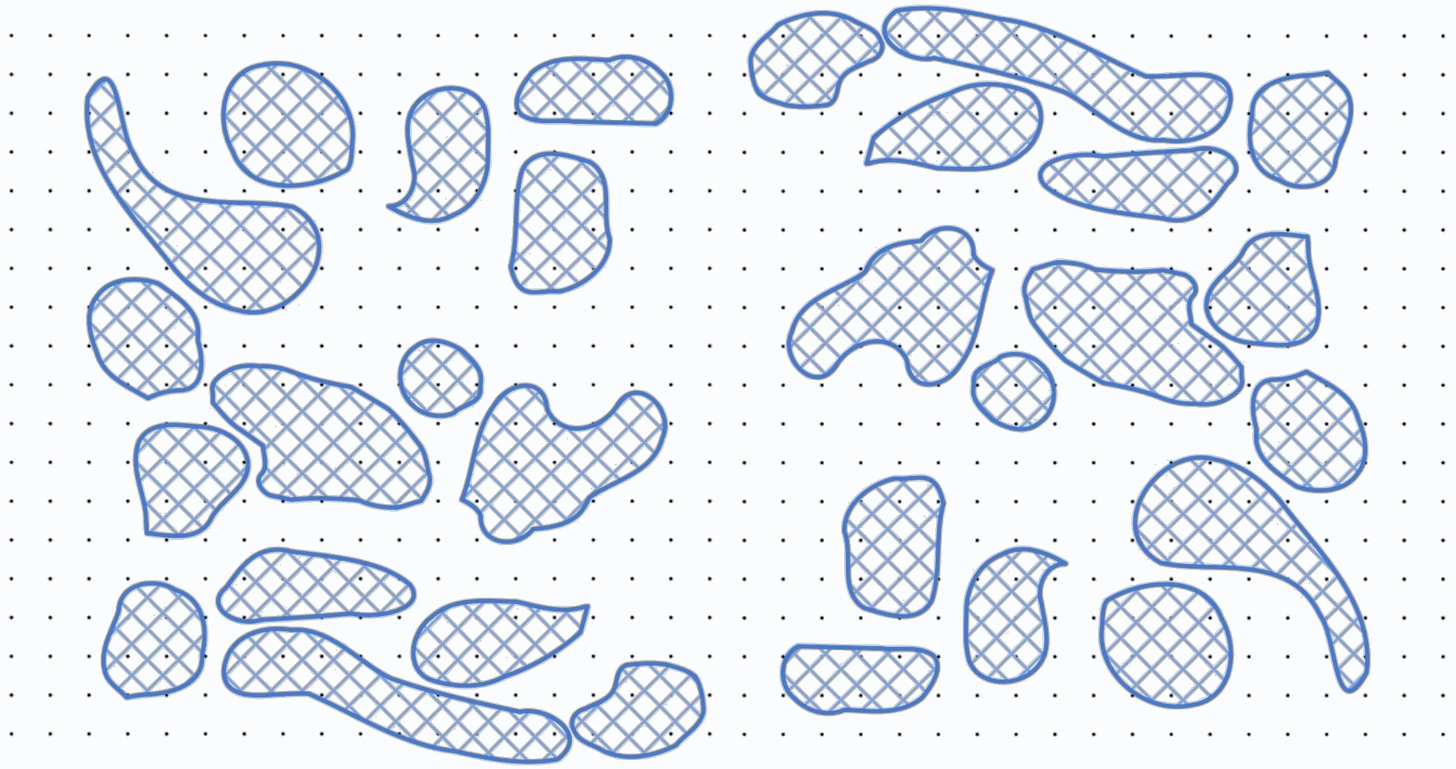
Partially phased

Nitrogenous bases ?

Do these methods work well in more difficult cases?

Phosphodiester backbone ?

Is there nucleic acid? If so, where?



Electron Density

$2mF_o$-DF_c

Partially phased

Nitrogenous bases ?





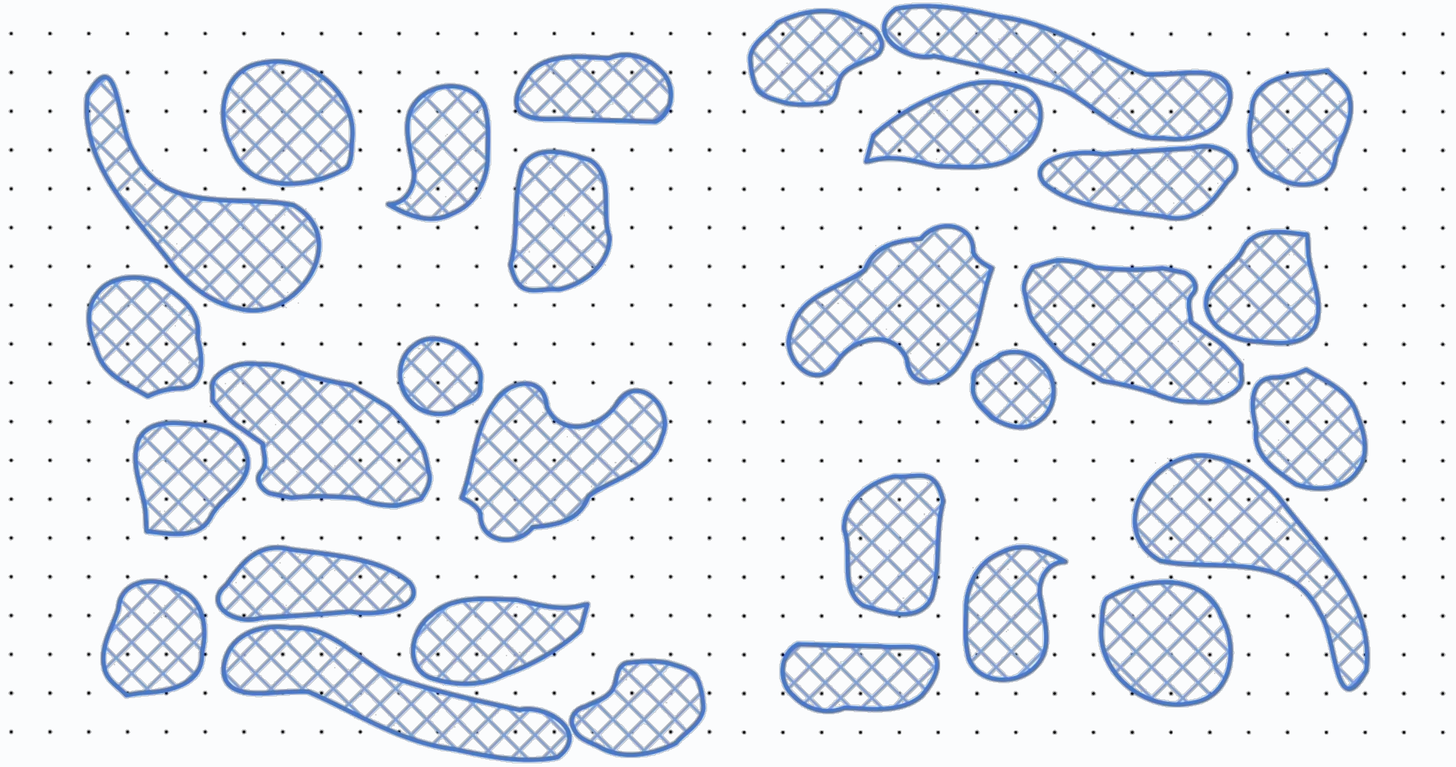


If they don’t, what happens now?

Is there nucleic acid?
If so, where?



Experienced Researcher



Electron Density

$2mF_o$-DF_c

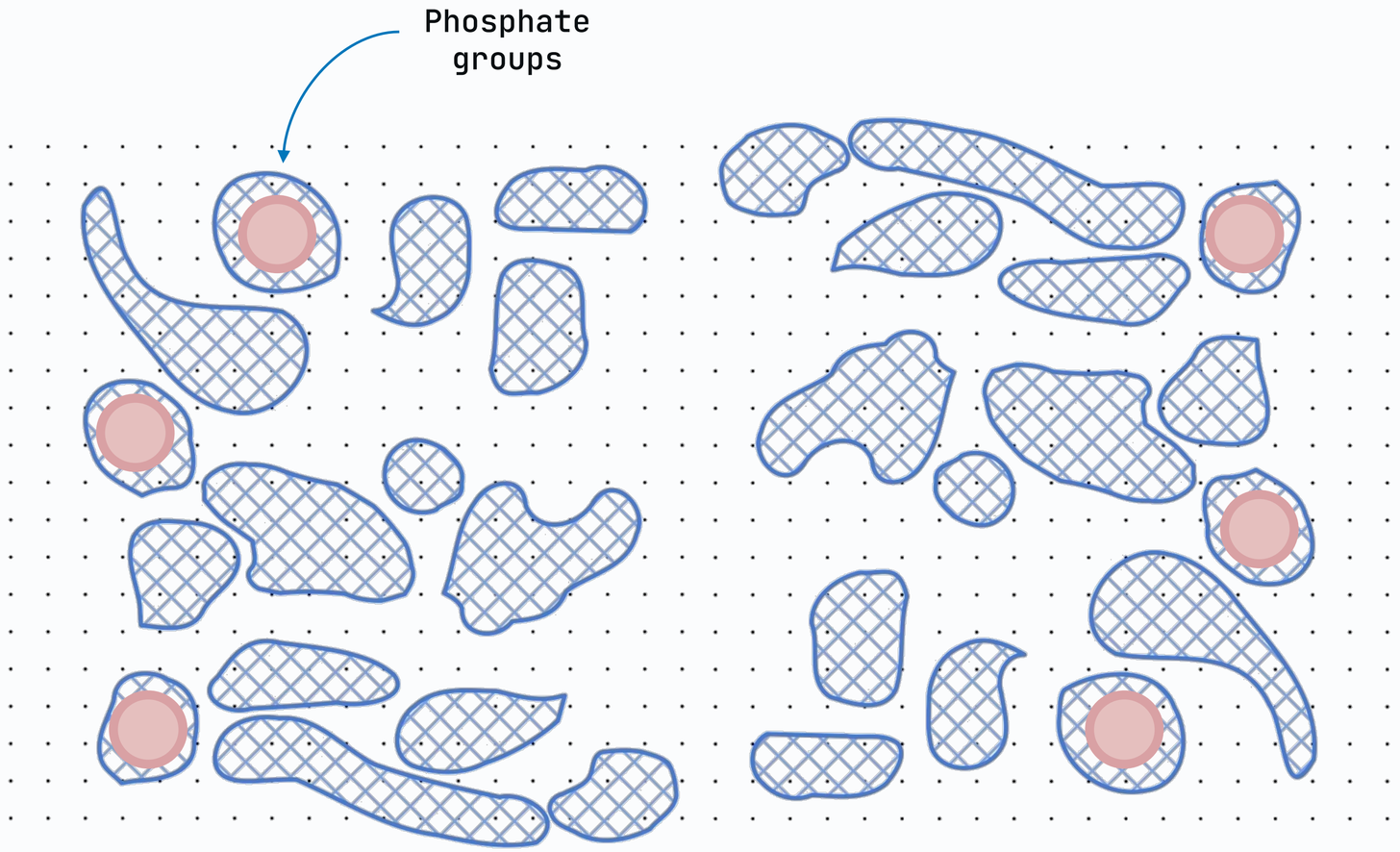
Partially phased

If they don’t, what happens now?

Is there nucleic acid?
If so, where?



Experienced Researcher



Electron Density

$2mF_o$-DF_c

Partially phased

If this is the reality, can we think of a better way?



Experienced Researcher

Offload hard work to a computer



Software Method

If this is the reality, can we think of a better way?



Experienced Researcher

Offload hard work to a computer



Software Method

Aim of existing packages







What’s so good about a human?

Strong pattern recognition skills

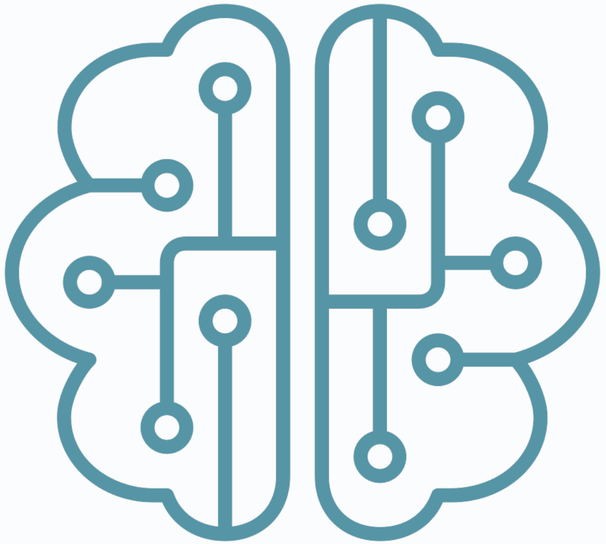


Experienced Researcher

Trained by looking at many examples
of their years of experience

What’s so good about a human?

Strong pattern recognition skills



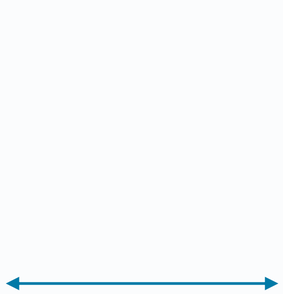
Artificial Intelligence Method

Trained by looking at many examples
of their years of experience

Can we replicate the skill of an experienced researcher with artificial intelligence?



Experienced Researcher





Software Method

Can we replicate the skill of an experienced researcher with artificial intelligence?



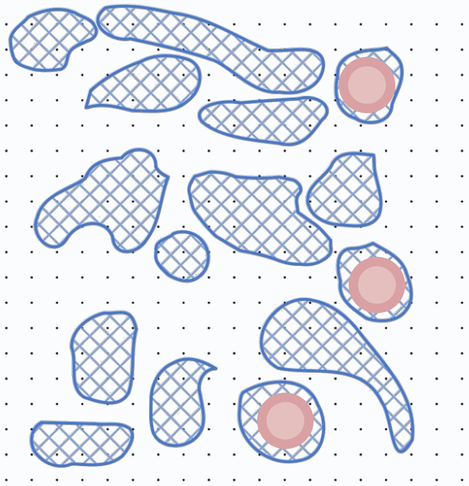
NucleoFind

Automated model building software package for nucleic acids

What is the aim of NucleoFind?

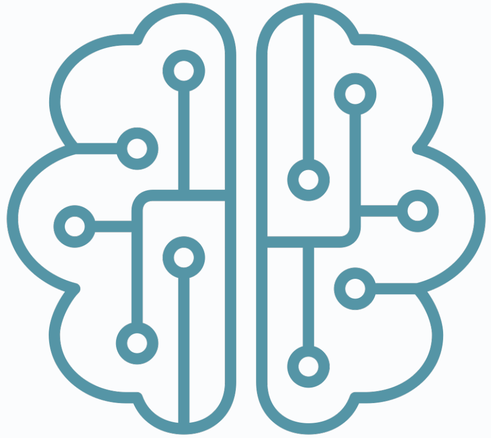


Experienced Researcher

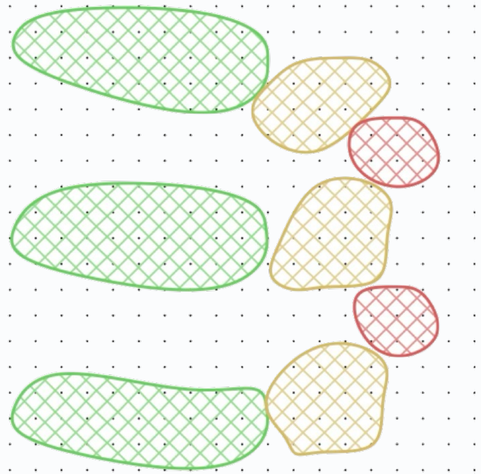


Manual Atomic Group Locations

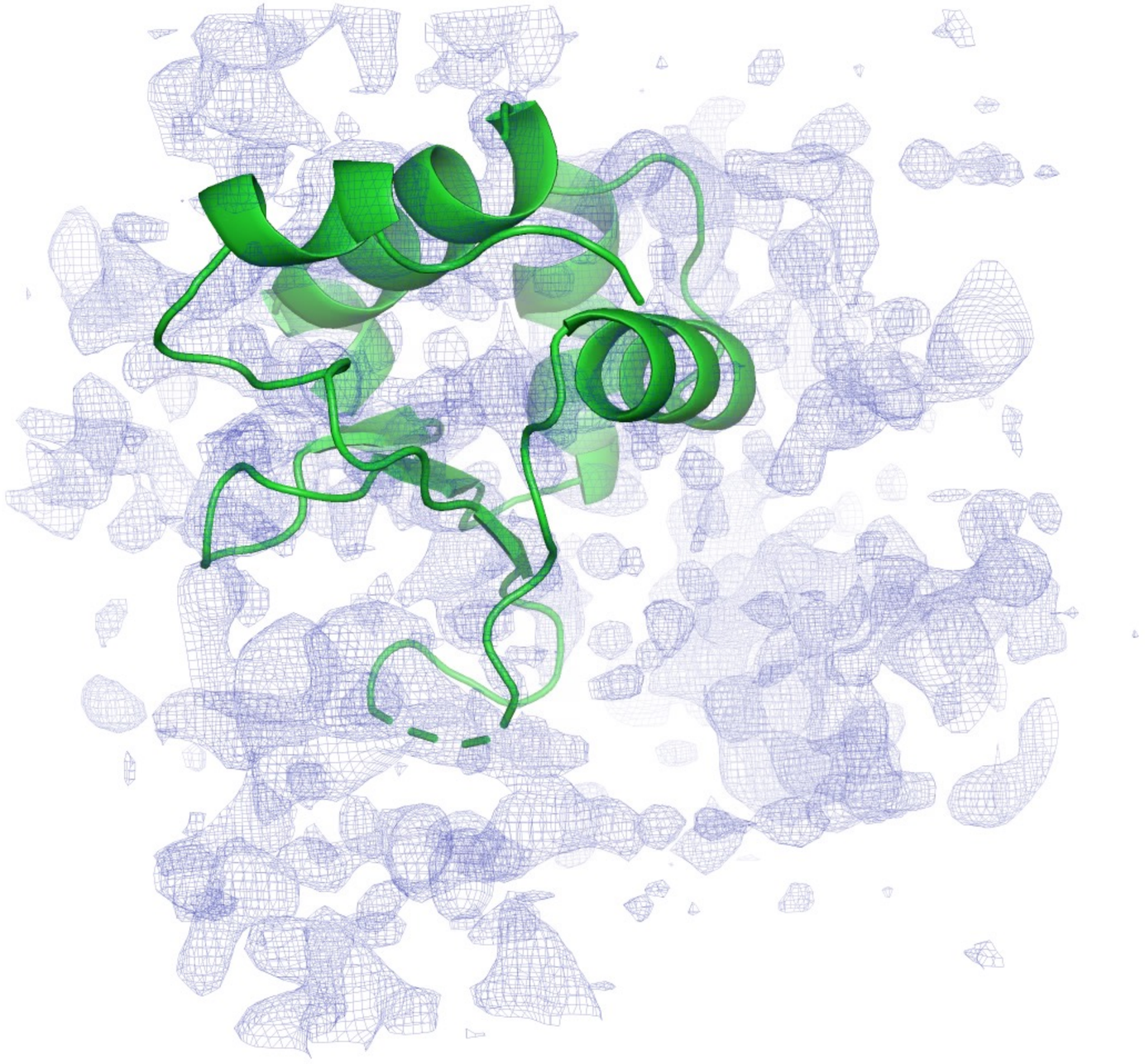
NucleoFind

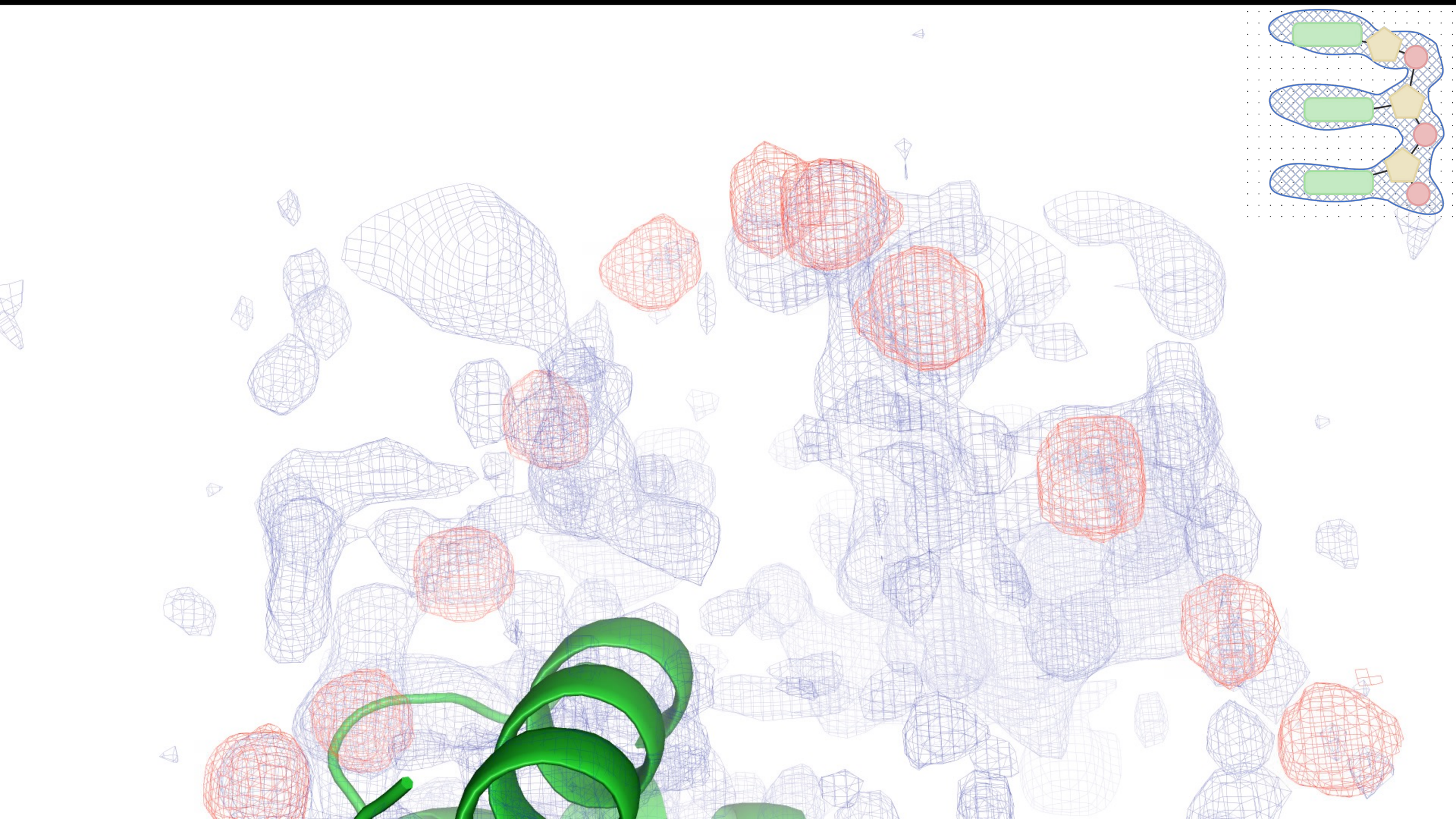


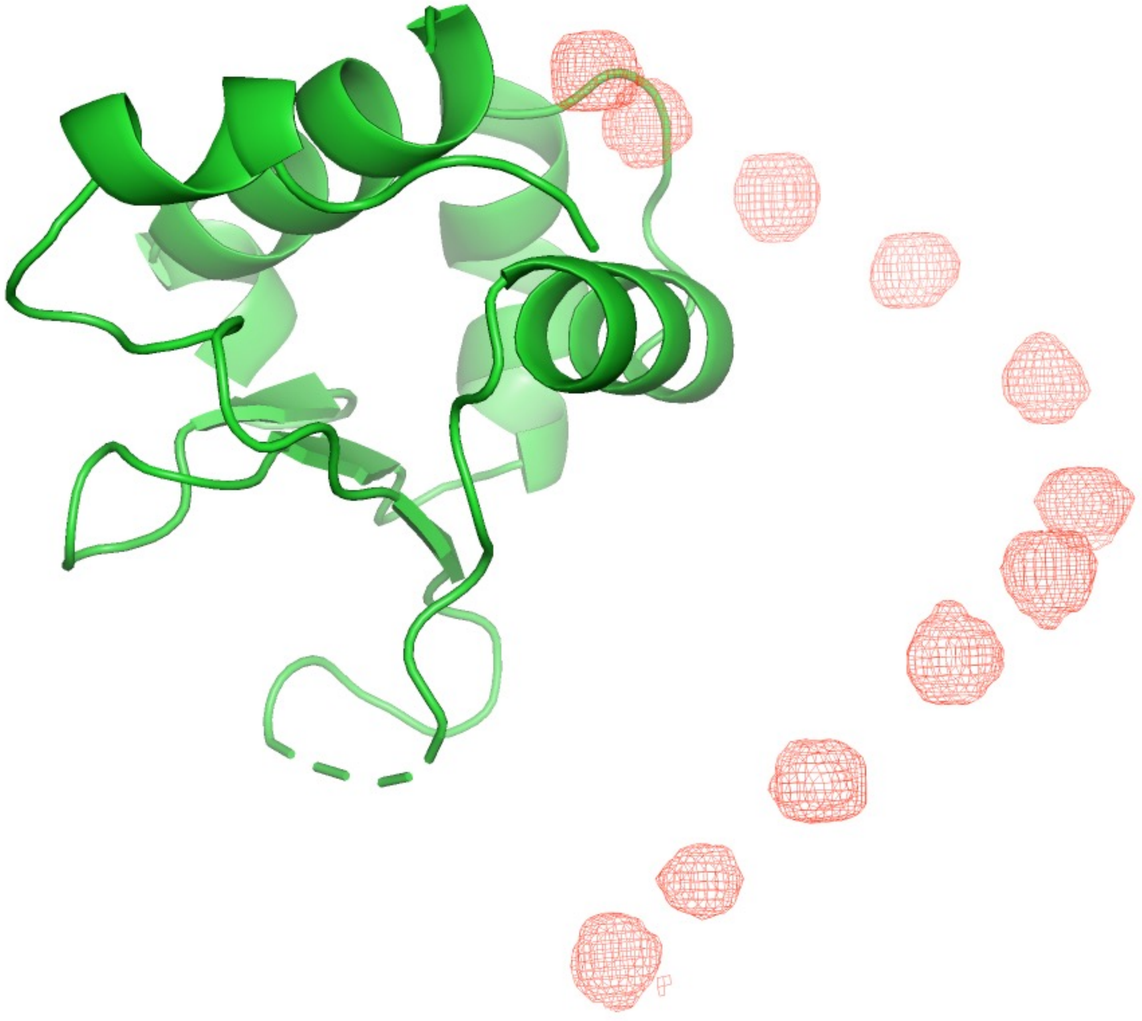
Model

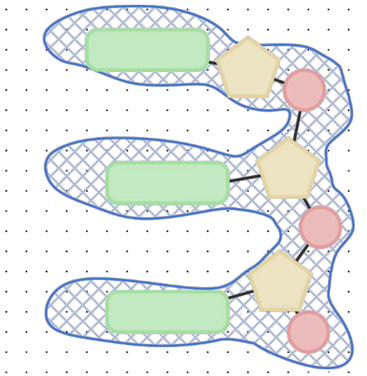


Predicted Atomic Group Locations

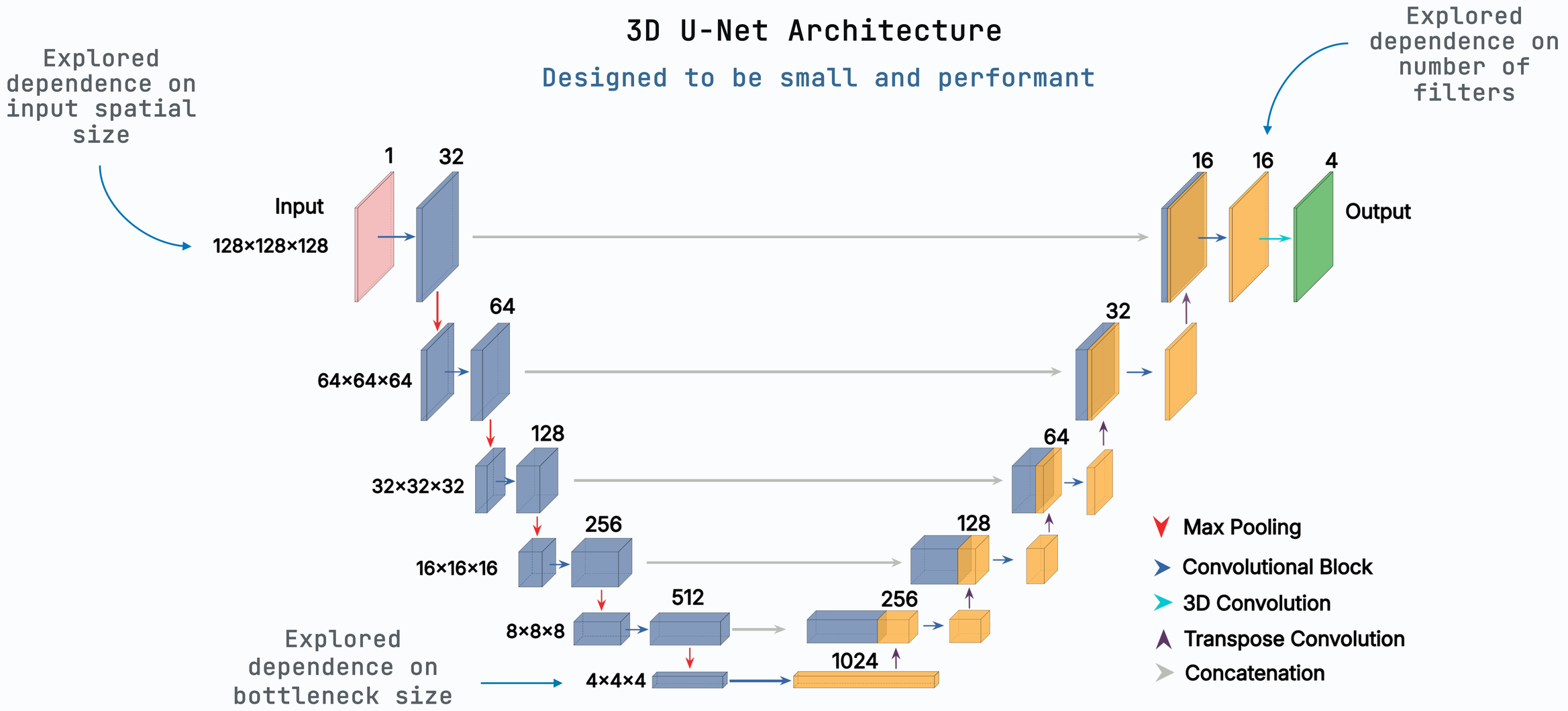






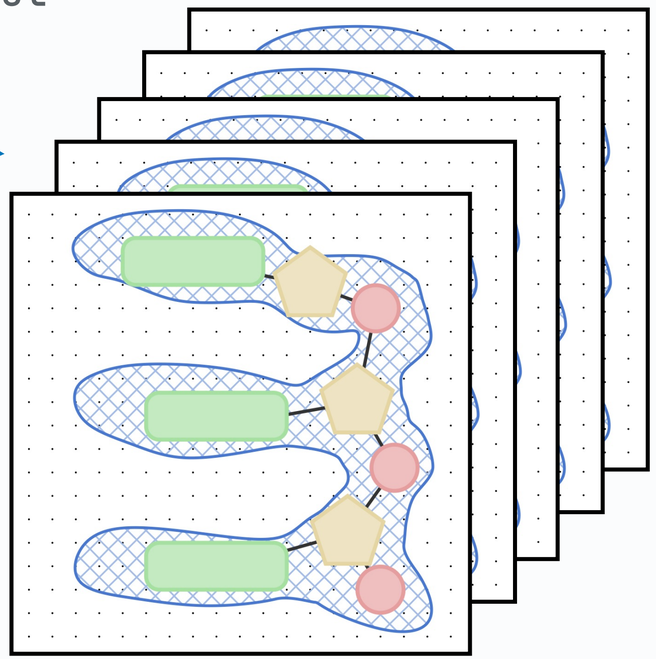


What is the model that powers NucleoFind?



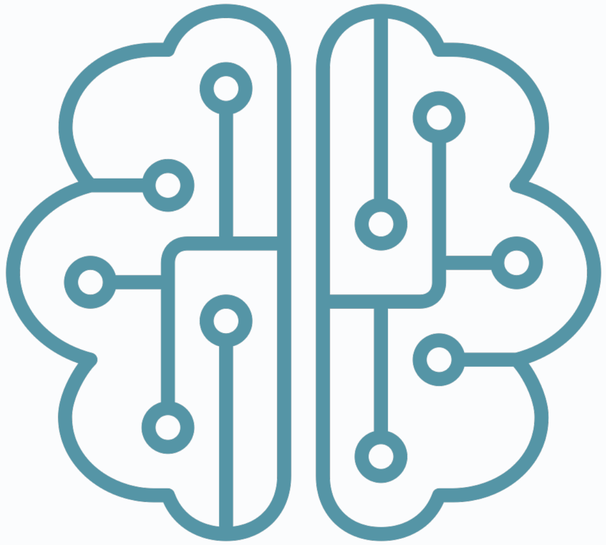
How was NucleoFind trained?

Ground truth generated from an atomic model



Many Examples

Mimics the experience of a researcher

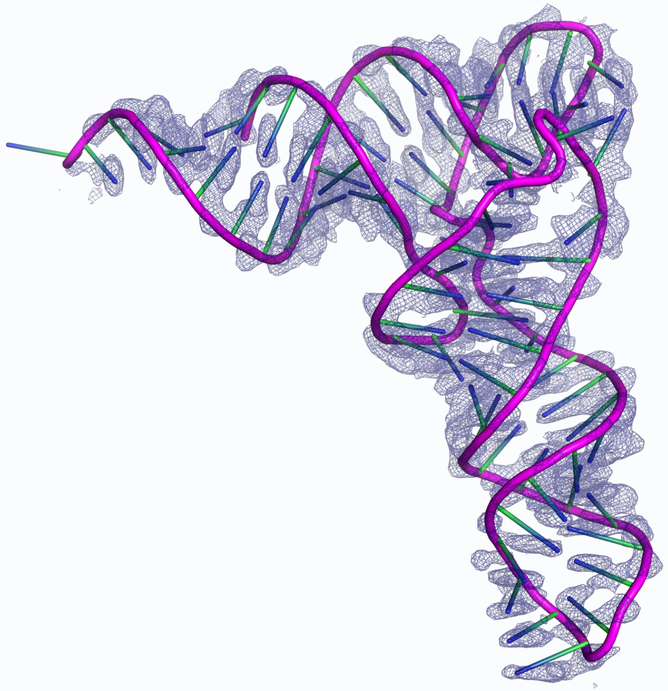


Trained Model

How was NucleoFind trained?

Nucleic Acid Only Maps

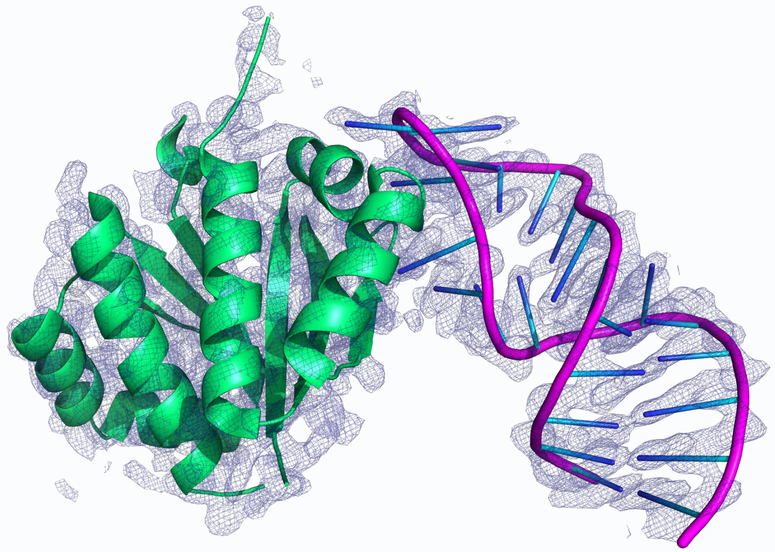
X-ray Crystallography
Cryo-EM



PDB Code - 1EHK

Protein-Nucleic Acid Maps

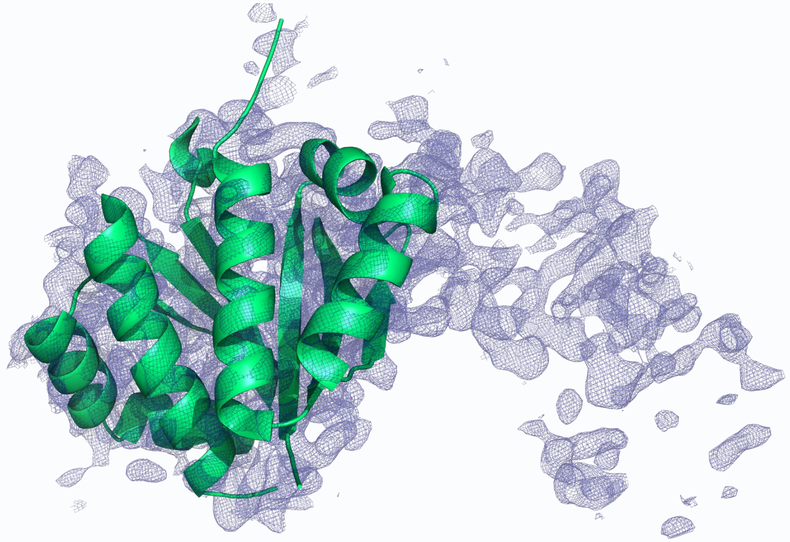
X-ray Crystallography
Cryo-EM



PDB Code – 2ANR

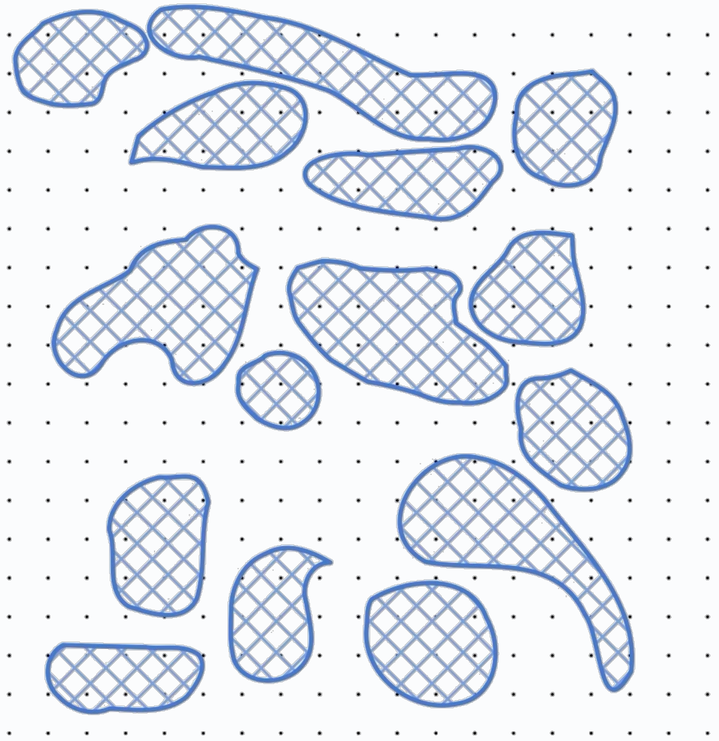
MR-like Protein-Nucleic Acid Maps

X-ray Crystallography



PDB Code – 2ANR

What is the model inside NucleoFind useful for?

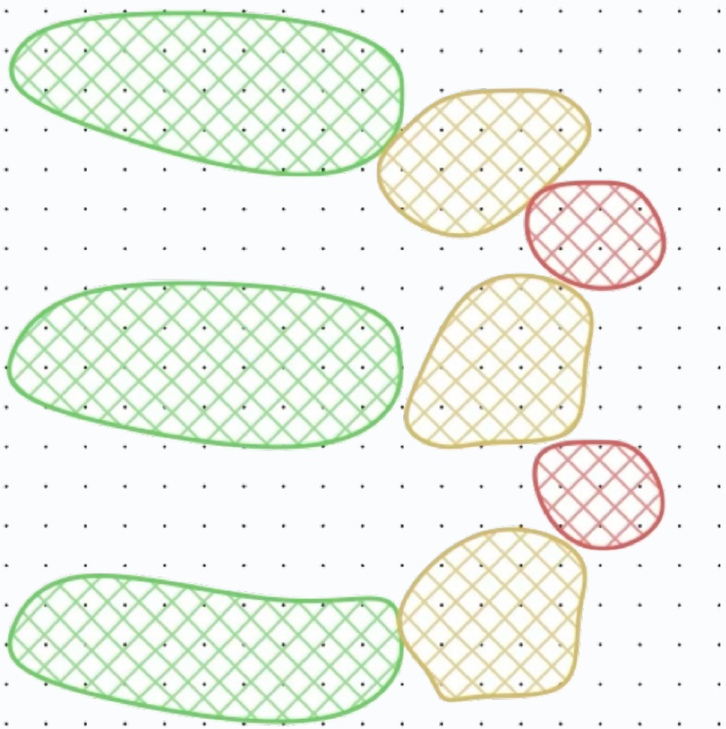


New Experimental Density

Is there nucleic acid?
If so, where?

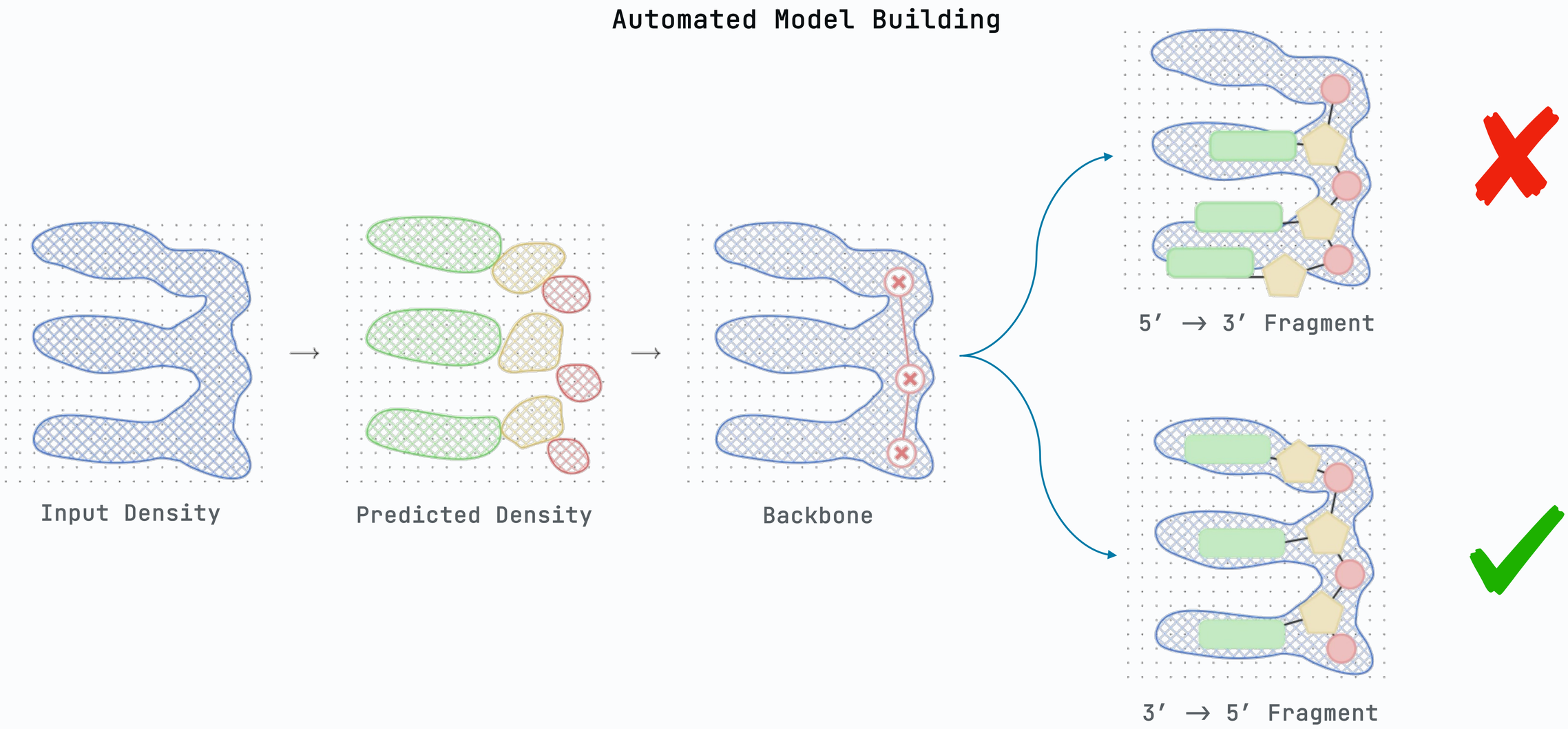
NucleoFind





Predicted Atomic Group Locations

So NucleoFind helps answer the question, what now?



NucleoFind is contained within the ModelCraft package



ModelCraft

START
Shift-field refinement
Sheetbend

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot

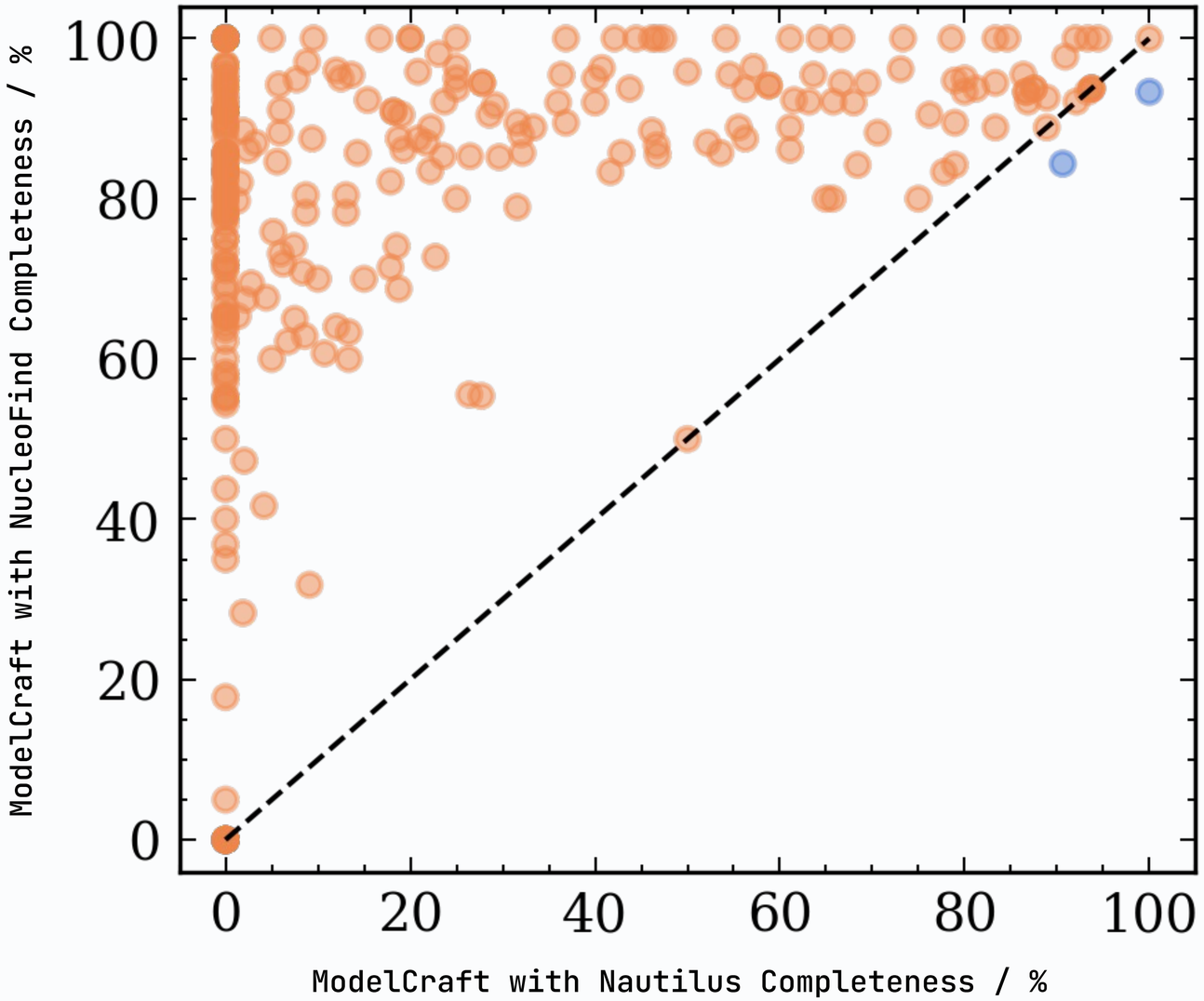
Repeated up to 25 times

END
Rebuild side chains
Coot

NucleoFind is a direct replacement for the existing nucleic acid model building program Nautilus

NucleoFind is a big improvement over the older method in ModelCraft

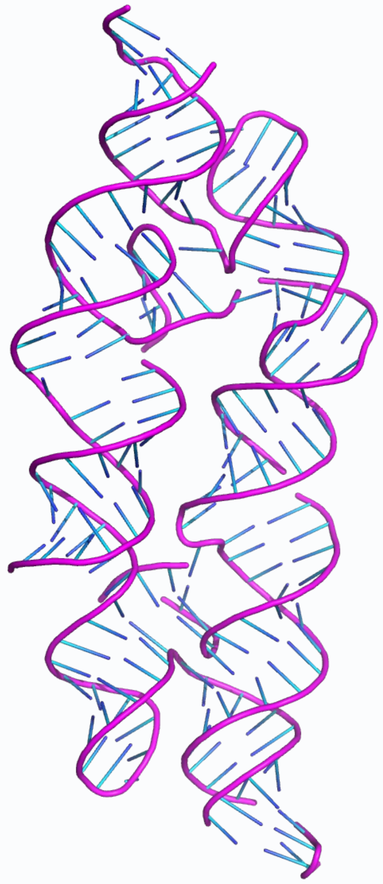
Test set of 288 protein-nucleic acid molecular replacement examples



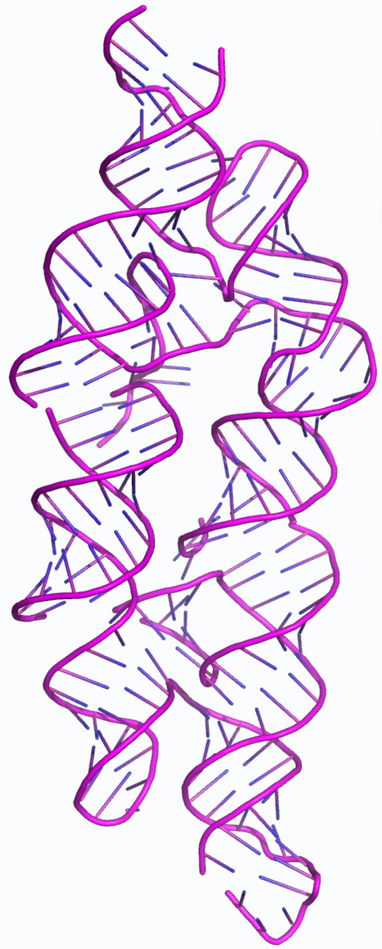
ModelCraft with NucleoFind produces a 55.1 pp increase in completeness over ModelCraft with Nautilus

NucleoFind models experimentally phased structures well

86 % complete in 25 minutes of automated processing time



Novel butterfly like double pseudoknot fold



NucleoFind has never seen a map phased with SAD

Phased with SAD, solved at 2.8 Å

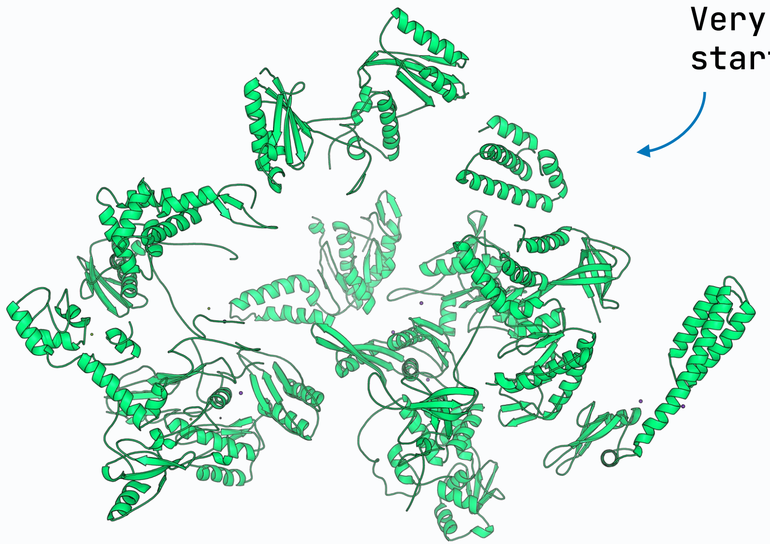
ModelCraft with NucleoFind Model

Deposited Model

Rous sarcoma virus frame-shifting RNA

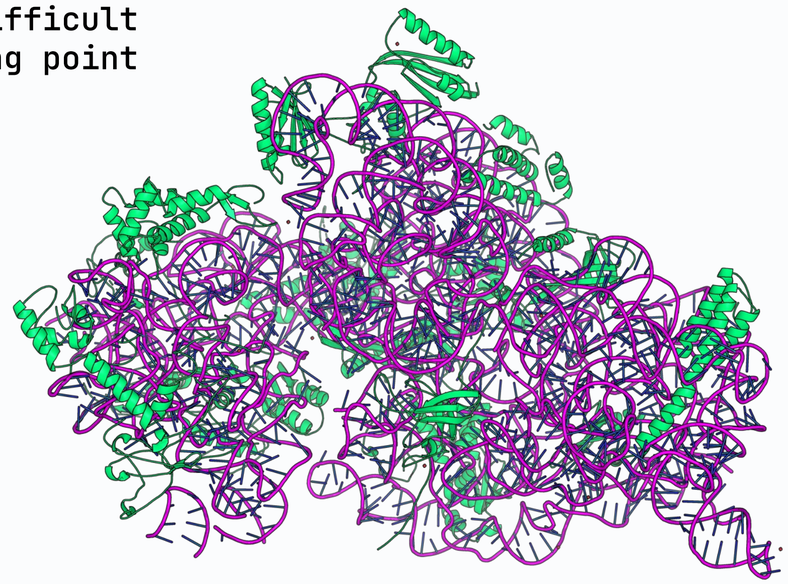
PDB Code - 9DID

NucleoFind models large protein-nucleic acid complexes well

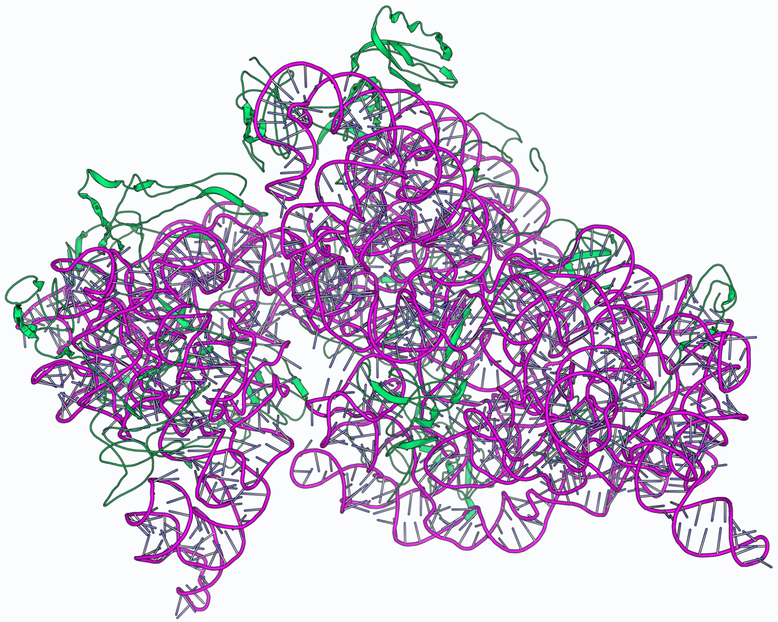


Very difficult
starting point

Molecular Replacement Solution



ModelCraft with NucleoFind Model



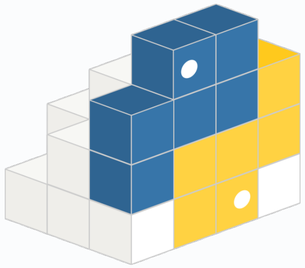
Deposited Model

88 % complete in 8
hours of automated
processing time

Thermus thermophilus 30S ribosomal subunit

PDB Code - 1IBK

How can I try NucleoFind and is it fast?



PyPi

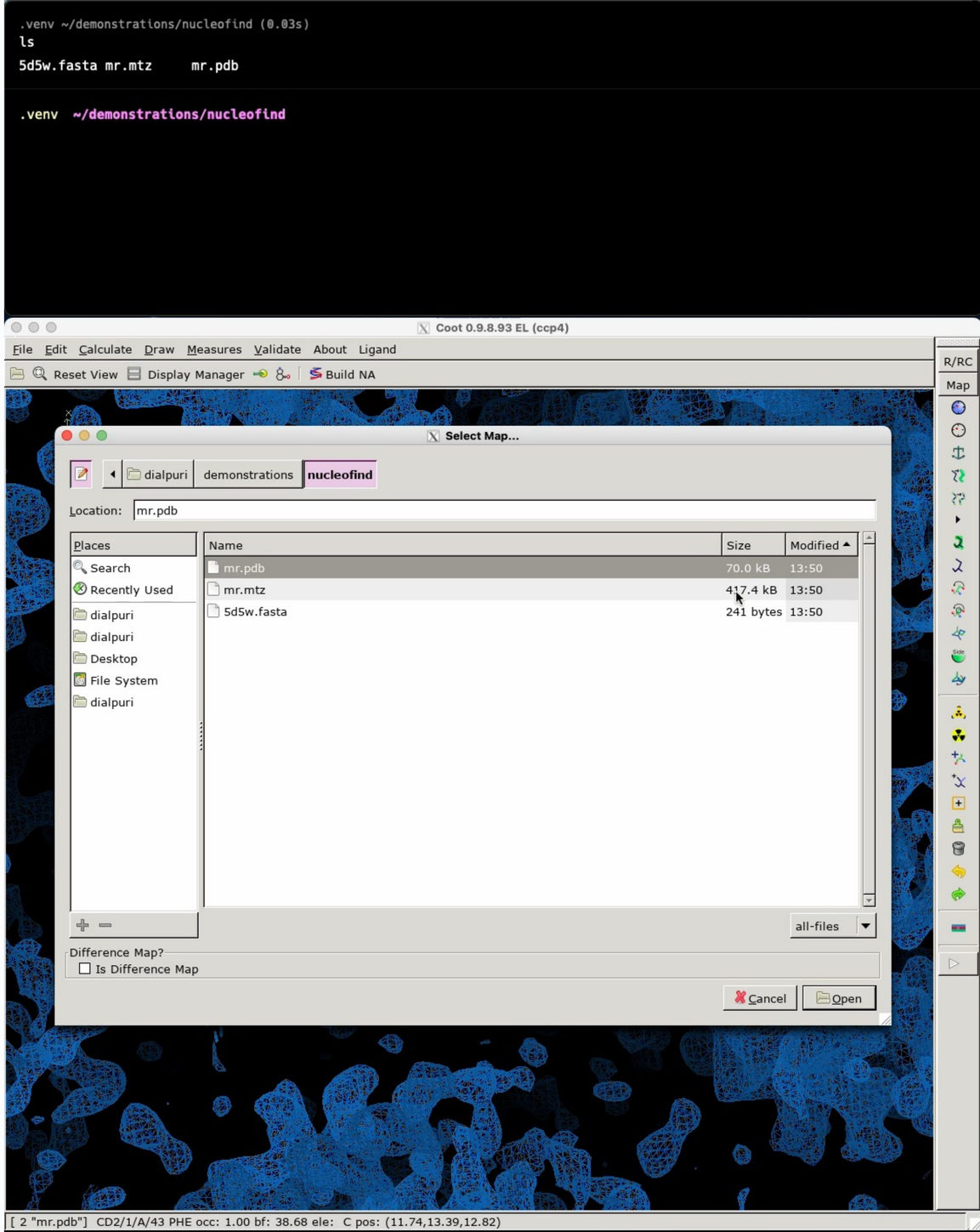
```
pip install nucleofind
```

CCP4

Latest CCP4 release



github.com/Dialpuri/NucleoFind



Is it just nucleic acid we can do this with?

github.com/Dialpuri/Sails



Deposited model to the PDB - no carbohydrates



Predicted carbohydrate positions from electron density



Automated carbohydrate model building results

Conclusions



NucleoFind

Automated model building software package for nucleic acids



ModelCraft

ModelCraft with NucleoFind is a powerful nucleic acid model building package



Sails

Automated model building software package for carbohydrates

BAY AREA LIGHT SOURCES JOINT USERS' MEETING

Upcoming features in Coot 1.4

Jordan Dialpuri, PhD - MRC Laboratory of Molecular Biology

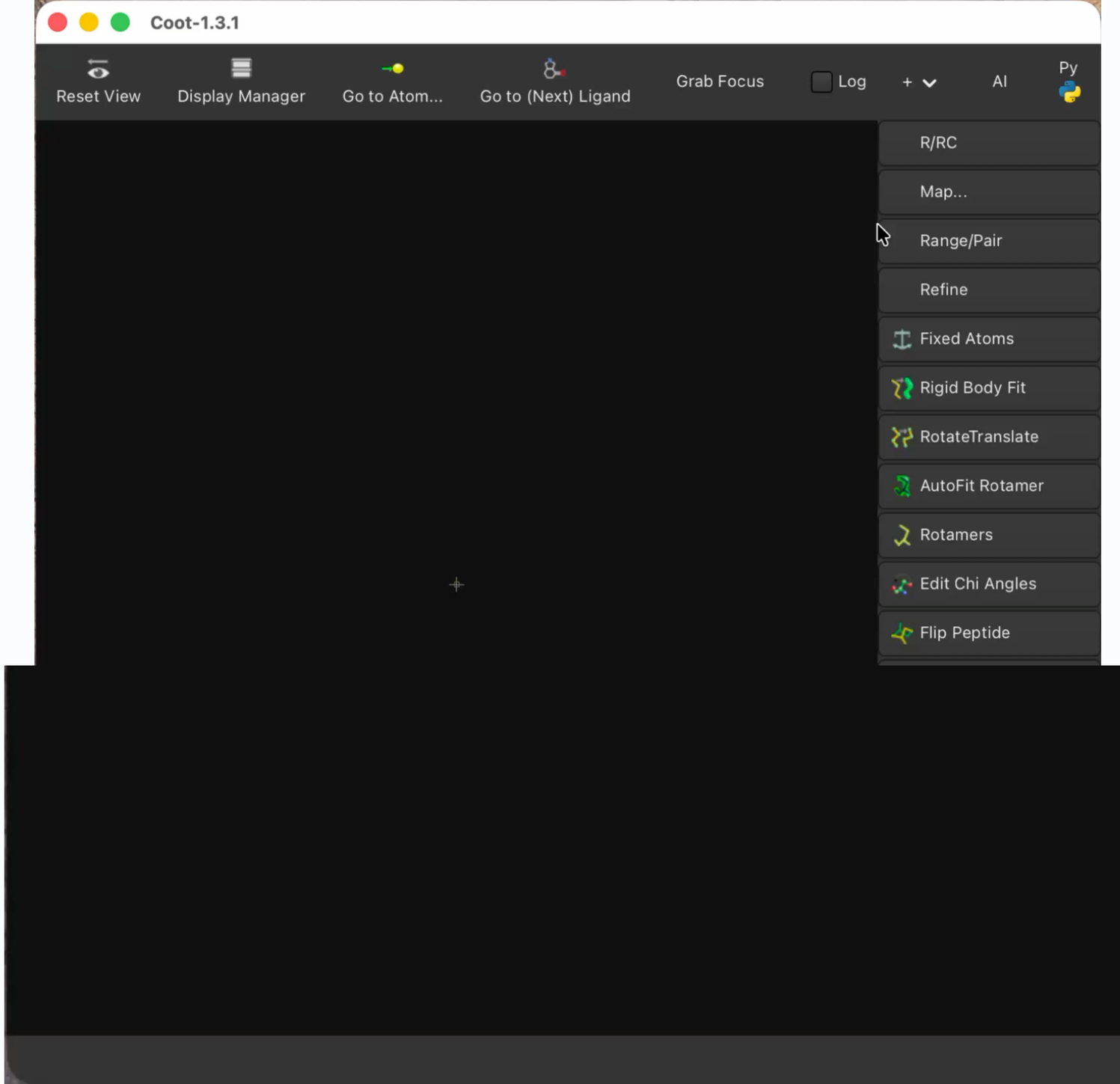
25th September 2026

What are the new features in Coot 1.4?

Coot Commands

PyMol-esque commands which control Coot

Can significantly speed up interactive workflows



What Coot commands can I use?

Model Editing

add solvent | ligand | water here

add altconf | OXT | terminal residue

autofit | backrub rotamer

pepflip

replace residue with ALA

delete residue | model | map | all

Refinement

refine chain | residue | sphere

refine B-factors

Display and Navigation

colour map blue

contour map 1 to 0.2

show | hide map | model

show | hide ribbons | surface | environment

center at x y z

go to A/1

next | prev

Fetching

fetch pdb | monomer (and map)

fetch pdb from pdb-redo

load /path/to/file/*.cif

What about beyond Coot 1.4?

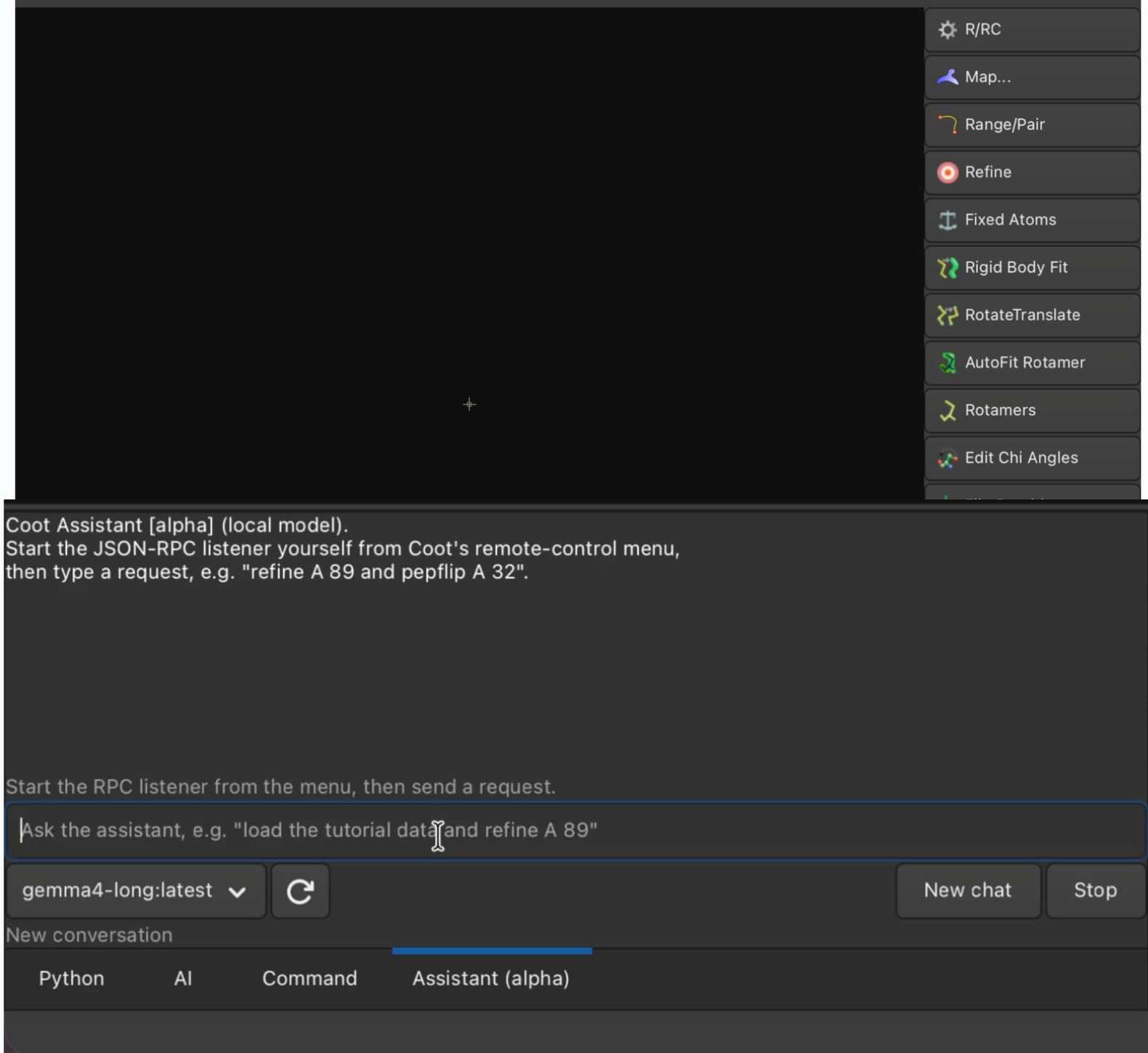
Coot Assistant

Using a local LLM - gemma4

Relies on new Coot commands

Uses pretrained embedding model to filter potential Coot tools

ALPHA VERSION - UNRELEASED WORK



Acknowledgements

University of York

Kathryn Cowtan

Paul Bond

Stuart McNicholas

Soon Wen Hoh

Clément Dégut

Jon Agirre

Haroldas Bagdonas

Lucy Schofield

Tim Kirk

William Cook

MRC Laboratory of Molecular Biology

Garib Murshudov

Paul Emsley

Martin Malý

Fei Long

Lucrezia Catapano

Alexander Shtyrov

Ivan Clayson





